Characterization of Circumferential Flux Distributions in Parabolic Trough Receivers Using Adaptive Gaussian Representations

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Supplementary Information

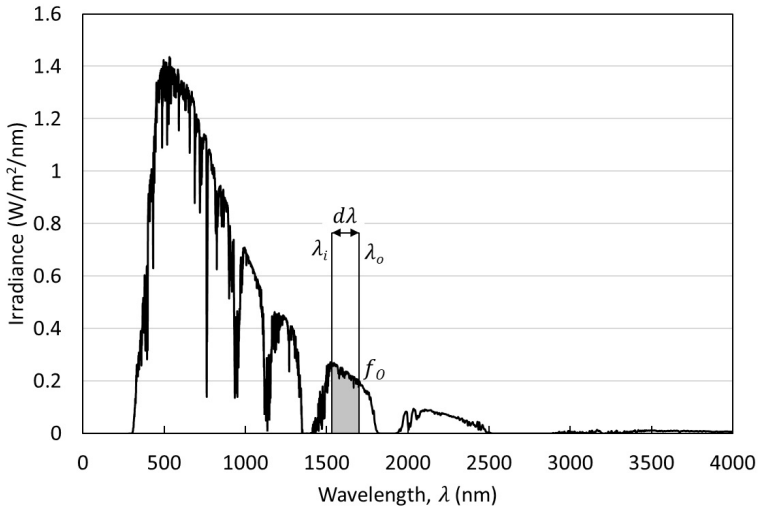


Figure S1: Reference solar spectral irradiance (ASTM G173-03, direct normal) used for the wavelength-resolved ray tracing.

Complete fitted parameter sets, the fitting code, and the ray-tracing outputs are available at `https://github.com/otsun-uib/GaussianLabSuppMat`. Amplitudes are reported as signed coefficients of each Gaussian component. For each candidate, the code reports whether the parameter covariance could be estimated, the largest relative standard error of the parameters (max. rel. s.e.), and whether the region is saturated ($n \leq k + 1$ for n points and k parameters, in which case AICc is undefined). Candidate fits with a component center more than 180° beyond either end of the receiver range 0°–180° (that is, outside −180° to 360°) or a component width above 180° are treated as non-physical and excluded before ranking.

Table S1: Candidate descriptors for Region 1 ($\varphi \leq 17.5°$, six points) of Jeter's 0° curve. The DoG fit is saturated (six parameters, six points) and its covariance could not be estimated. n/a: not defined

Model	R^2	RMSE	MAE	SSR	AIC	AICc	BIC	Max. rel. s.e. (%)	Note
Sup.G.	0.7423	0.8817	0.8384	4.665	6.5	46.5	5.7	5×10^5	
Inv.G.	0.9911	0.1638	0.1038	0.161	−13.7	26.3	−14.5	21	
DoG	0.9987	0.0627	0.0467	0.024	−21.2	n/a	−22.5	n/a	saturated
Tru.G.	0.7388	0.8877	0.8441	4.728	6.6	46.6	5.7	n/a	

Table S2: Candidate descriptors for Region 2 ($\varphi > 17.5°$, 53 points). GMM4 was excluded by the plausibility screen, and GMM3 was selected. Max. rel. s.e.: largest relative standard error of the parameters. n/a: not defined

Model	R^2	RMSE	MAE	SSR	AIC	AICc	BIC	Max. rel. s.e. (%)	Note
Gaussian	0.9511	3.7347	2.8547	739.258	145.7	146.2	151.6	3.5	= F.G.
Sup.G.	0.9831	2.1950	1.4627	255.350	91.3	92.2	99.2	6.2	
Skw.G.	0.9958	1.0919	0.8376	63.184	17.3	18.1	25.2	6.3	
Twn.G.	0.9882	1.8342	1.2759	178.307	76.3	78.1	88.1	17.5	= DoG
Inv.G.	0.9526	3.6755	2.9100	715.975	146.0	146.8	153.9	4.5	
DoG	0.9882	1.8342	1.2759	178.307	76.3	78.1	88.1	17.5	= Twn.G.
GMM3	0.9990	0.5418	0.4090	15.560	−47.0	−42.8	−29.2	56	selected
GMM4	0.9996	0.3193	0.2280	5.402	−97.0	−89.2	−73.4	1.4×10^6	excluded
Tru.G.	0.9543	3.6086	2.6988	690.165	144.0	144.9	151.9	n/a	
F.G.	0.9511	3.7347	2.8547	739.258	145.7	146.2	151.6	3.5	= Gaussian
Spt.G.	0.9953	1.1516	0.8772	70.283	23.0	23.8	30.8	4.5	

Table S3: Parameters and standard errors (± one standard error) of the selected GMM3 descriptor in Region 2 of Jeter's 0° curve. Standard errors are the square roots of the diagonal of the parameter covariance matrix returned by the least-squares fit. The selected Region 1 descriptor (DoG) is saturated, and its covariance could not be estimated.

Component	A	μ (°)	σ (°)
1	33.3 ± 0.8	26.6 ± 2.2	26.7 ± 8.0
2	38.3 ± 7.9	71.8 ± 1.6	17.5 ± 0.8
3	1.02 ± 0.38	165.0 ± 8.9	17.6 ± 9.9

Table S4: Sensitivity of the selected descriptors to the split angle φ_S for Jeter's 0° curve. n_1 is the number of points in Region 1, and "Sat." marks Region 1 fits with $n_1 \leq k + 1$. The digitized curve has no points between 14.8° and 21.0°, so splits of 15°–20° give identical regions, and the same holds for 22.5°–27.5°. The first row is the automatic estimate (12.533°, which includes the point at 12.53°), and the row at 12.53° uses the rounded angle and has one point fewer in Region 1.

φ_S (°)	n_1	Region 1	Region 2	k	R^2	RMSE	AIC	BIC	Sat.
12.533 (automatic)	5	Inv.G.	GMM3	13	0.9986	0.596	−35.1	−8.1	yes
12.53	4	Inv.G.	GMM4	16	0.9997	0.294	−112.5	−79.2	yes
15, 17.5, 20	6	DoG	GMM3	15	0.9990	0.514	−48.5	−17.4	yes
22.5, 25, 27.5	7	Inv.G.	GMM4	16	0.9997	0.277	−119.6	−86.3	no
30	8	Two-G.	GMM4	18	0.9998	0.232	−136.4	−99.0	no
37.5	9	GMM3	GMM4	21	0.9990	0.500	−39.9	3.8	yes
45	10	GMM3	DoG	15	0.9990	0.513	−48.8	−17.6	yes
67.5	17	GMM4	GMM3	21	0.9999	0.161	−173.2	−129.6	no

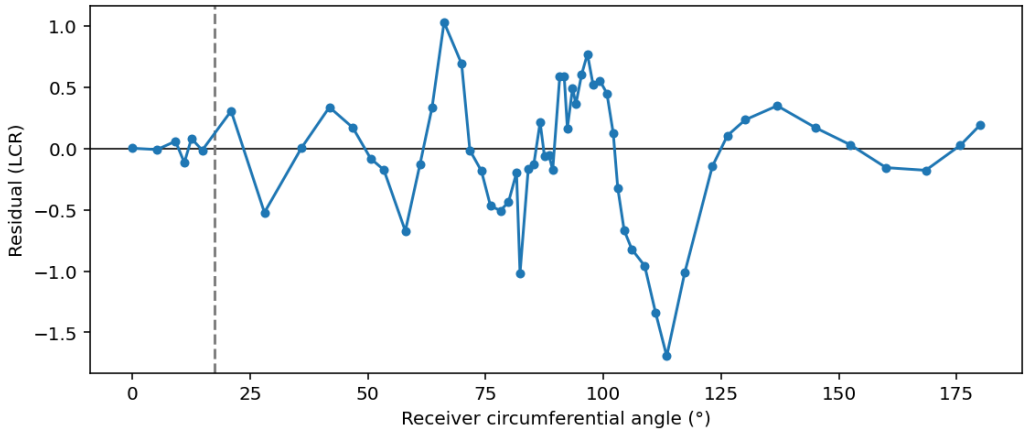


Figure S2: Residuals (data minus fit) for Jeter's 0° curve with the DoG (Region 1) and GMM3 (Region 2) descriptors. The dashed line marks the split angle $\varphi_S = 17.5°$.

Table S5: Shadowing-region Inv.G. fits used to compare the baseline B_L: region boundary, center μ, baseline B_L, ray-traced LCR at the first angular point, and optical efficiency η_o. Materials are wavelength-dependent unless marked (c) for wavelength-independent.

Sweep	Case	Region 1 (°)	μ (°)	B_L	LCR (first point)	η_o
Height	$F - 2.0(D_R/2)$	0–22.5	1	8.3	8.34	0.266
	$F - 1.5(D_R/2)$	0–27.5	0	11.0	11.37	0.386
	$F - 1.0(D_R/2)$	0–27.5	0	14.9	15.31	0.649
	$F - 0.5(D_R/2)$	0–22.5	3	22.65	22.66	0.784
	F	0–17.5	4.2	34.9	35.00	0.795
Diameter	70 mm	0–17.5	3	34.8	35.09	0.797
	80 mm	0–22.5	0	28.0	28.13	0.802
	90 mm	0–15	0	24.2	24.55	0.806
	100 mm	0–22.5	0	20.5	20.97	0.810
	110 mm	0–22.5	0	18.38	18.69	0.815
	120 mm	0–22.5	0	16.5	17.03	0.818
CSR	0.0	0–17.5	3	28.2	28.27	0.816
	0.05	0–16	3	29.4	29.52	0.806
	0.1	0–12.5	3	29.0	29.31	0.802
	0.2	0–12.5	3	28.4	28.27	0.792
Rim angle	70°	0–17.5	1	34.5	34.68	0.797
	80°	0–17.5	3	34.8	35.12	0.797
	90°	0–22.5	0	35.2	35.33	0.794
	100°	0–27.5	0	35.0	34.97	0.789
	110°	0–27.5	0	36.0	36.42	0.779
	120°	0–27.5	0	37.0	36.84	0.763
	130°	0–37.5	3	37.5	37.58	0.725
Materials	Bare, CSR 0	0–17.5	3	37.5	37.73	0.859
	Bare, CSR 0.05	0–17.5	3	37.0	37.27	0.847
	Glass tube, CSR 0.05	0–12.5	2.8	34.8	35.01	0.778
	Glass with coating, CSR 0.05	0–17.5	2.8	33.8	33.90	0.797
	Bare, CSR 0.05 (c)	0–22.5	4	33.5	33.53	0.881
	Glass tube, CSR 0.05 (c)	0–22.5	3	28.0	28.19	0.804
	Bare, CSR 0 (c)	0–12.5	1	31.0	31.14	0.887

Table S6: Descriptors and angular ranges (°) of the curves plotted in the parametric-study figures. Ranges are listed in the order of the descriptors. D_R is the receiver diameter (mm).

Case	Descriptor(s)	Range(s) (°)
CSR sweep		
CSR 0.0	Inv.G. + GMM4	0–17.5 / 17.5–180
CSR 0.05	Inv.G. + GMM4	0–16 / 16–180
CSR 0.1	Inv.G. + GMM4	0–12.5 / 12.5–180
CSR 0.2	Inv.G. + GMM4	0–12.5 / 12.5–180
CSR 0.5	GMM4	0–180
CSR 0.7	GMM3	0–180
Materials (a): wavelength-independent		
Bare receiver, CSR 0	Inv.G. + Inv.G. + Skw.G.	0–12.5 / 12.5–77.5 / 77.5–180
Bare receiver, CSR 0.05	Inv.G. + GMM4	0–22.5 / 22.5–180
Glass tube, CSR 0.05	Inv.G. + GMM3	0–22.5 / 22.5–180
Altarawneh et al., "point source"	Inv.G. + Inv.G. + Skw.G.	0–12.5 / 12.5–72.5 / 72.5–180
Altarawneh et al., sun shape	Inv.G. + GMM4	0–20 / 20–180
Jeter, sun shape	DoG + GMM3	0–15 / 15–180
Materials (b): wavelength-dependent		
Bare receiver, CSR 0	Inv.G. + GMM4	0–17.5 / 17.5–180
Bare receiver, CSR 0.05	Inv.G. + GMM3	0–17.5 / 17.5–180
Glass tube, CSR 0.05	Inv.G. + GMM3	0–12.5 / 12.5–180
Glass with coating, CSR 0.05	Inv.G. + GMM3	0–17.5 / 17.5–180
Rodriguez-Sanchez & Rosengarten	Inv.G. + GMM4	0–12.5 / 12.5–180
Receiver diameter		
$D_R = 50, 60$	Spt.G.	0–180
$D_R = 70$	Inv.G. + GMM3	0–17.5 / 17.5–180
$D_R = 80$	Inv.G. + GMM3	0–22.5 / 22.5–180
$D_R = 90$	Inv.G. + GMM3	0–15 / 15–180
$D_R = 100, 110, 120$	Inv.G. + GMM4	0–22.5 / 22.5–180
Receiver height (a): below focus		
$F - 2.0(D_R/2)$	Inv.G. + GMM3	0–22.5 / 22.5–180
$F - 1.5(D_R/2)$, $F - 1.0(D_R/2)$	Inv.G. + GMM3	0–27.5 / 27.5–180
$F - 0.5(D_R/2)$	Inv.G. + GMM3	0–22.5 / 22.5–180
F	Inv.G. + GMM3	0–17.5 / 17.5–180
Receiver height (b): above focus		
$F + 0.5(D_R/2)$	Sup.G.	0–180
$F + 1.0(D_R/2)$	Twn.G.	0–180
$F + 1.5(D_R/2)$, $F + 2.0(D_R/2)$	Sup.G.	0–180
Rim angle (a)		
$\Psi = 10, 20, 30$	Sup.G.	0–180
$\Psi = 40, 50, 60$	Spt.G.	0–180

Case	Descriptor(s)	Range(s) (°)
Ψ = 70, 80	Inv.G. + GMM3	0–17.5 / 17.5–180
Rim angle (b)		
Ψ = 90	Inv.G. + GMM3	0–22.5 / 22.5–180
Ψ = 100, 110, 120	Inv.G. + GMM3	0–27.5 / 27.5–180
Ψ = 130	Inv.G. + GMM3	0–37.5 / 37.5–180
Ψ = 140, 150	GMM3	0–180
Asymmetric flux ($\theta = 1°$): receiver diameter, full circumference		
D_R = 50	GMM4 + Gaussian	−180–97.5 / 97.5–180
D_R = 60	GMM4 + Gaussian	−180–92.5 / 92.5–180
D_R = 70	GMM3 + Gaussian	−180–97.5 / 97.5–180
D_R = 80	GMM4 + Gaussian	−180–97.5 / 97.5–180
D_R = 90	GMM4 + Gaussian	−180–92.5 / 92.5–180
D_R = 100	GMM4 + GMM4	−180– −37.5 / −37.5–180
D_R = 110	GMM4 + GMM4	−180– −32.5 / −32.5–180
D_R = 120	GMM4 + GMM4	−180– −30 / −30–180